Short Paper

Different Exemplars, Close Together: How the Arrangement of Training Text Teaches Small Transformers Which Features Persist

Jon-Paul Cacioli

Independent researcher, Melbourne, Australia
synthium@hotmail.com

We show that the arrangement of training text, with content held fixed, decides which feature small transformers expect to persist. We tested this with a regularity that holds across items: a learner that sees one new object called a dax *might expect the next* dax *to share its shape rather than its colour. In Study 1 we trained small transformers on artificial languages, arranging every pass afresh so that no text was seen twice. In random order the models expected the shape to change (shape-over-colour index −1.56 nats); grouped by kind, the same sentences taught them to expect it to stay (9.52). The effect required a feature fixed within kinds, appeared at moderate clustering, and was more selective when nearby sentences described different objects of a kind rather than one object again. When a kind always had one shape, the models barely tied the repetition of shape to the noun; when a kind's shape changed between stretches of text, they did so for trained nouns about as strongly as the text warranted. Models trained on one fixed file memorised it; their effects differed in size and, in random order, in sign. In Study 2, object labels were more clustered in speech around young children than in adult conversation or written text, though nearly half of the repetitions returned to the same object. In Study 3, a shape-over-colour expectation grew with Pythia model size, but no Pythia model preferred shape over texture. The models used the noun when they needed it.*

1. Introduction

What a language model learns depends on its training data. Most work on that data asks about content: which sources, how much, and how close to a child's input (Warstadt et al. 2023; Feng, Goodman, and Frank 2024). Less asks about arrangement, the order in which the same content is presented, although every training pipeline sets it when it packs documents into windows, orders them or generates synthetic text (Shi et al. 2024; Zhao et al. 2024). Natural sources also differ in arrangement. A word that has appeared once in a document is far more likely to appear again soon than its frequency predicts (Church 2000), and parents' talk to toddlers returns to the same objects in clusters (Slone et al. 2023). We ask what arrangement alone, with content held fixed, changes in what a next-word learner extracts from its data.

We study a regularity that holds across items rather than within any one of them. A child who hears "this is a dax" about a new object will call other objects "dax" if they share its shape, even when they differ in size or texture (Landau, Smith, and Jones 1988).

© 2026 Association for Computational Linguistics

The child has learned a fact about kinds in general: things with the same name tend to share a shape. Kemp, Perfors, and Tenenbaum (2007) called this an overhypothesis, after Goodman (1955).

We give a model one description of an object of a new kind, such as *A snerg is a krannish pulek skorli thing*, where the three adjectives name a shape, a colour and a texture. We then compare how strongly the model expects a second snerg to keep the shape with how strongly it expects it to keep the colour. The difference, in nats (natural-log units of probability), is our index. A positive index means the model expects shape to stay the same more than colour.

A positive index can come from two different expectations. One is tied to the kind: objects with the same name share a shape. The other is not: whatever object is described next will keep the shape of the last one. Only the first is an overhypothesis about named kinds. We separate them by changing the noun at test, which needs nouns the model can track. If the second object has a different name, the first expectation predicts a smaller preference, and the second predicts the same preference as before. The paradigm is therefore first a test of which feature a model expects to persist, and a test of an overhypothesis about named kinds only if the noun matters.

We ask four questions in three studies. Study 1 trains small transformers on artificial languages in which we control both content and order, and addresses the first three questions. Studies 2 and 3 address the fourth, with natural corpora and with pretrained models.

1. **Does grouping the sentences about each kind teach the fixed feature?** We predicted that the same sentences grouped into short runs by kind would give a higher index than in random order, and that random order alone would give an index above zero. We also predicted that the gain would need a feature fixed within kinds and would follow it, so that corpora with colour fixed would teach a preference for colour. Last, we predicted that the index would rise with measured clustering, and that clustering at the level we found for object labels in speech to children would already beat random order.

2. **What kind of grouping helps?** A longer training window holds more sentences about a kind without changing their order, and we predicted that it would raise the index. Nearby sentences can also describe a different object of the kind or the same object again. We predicted that different objects, which show what varies within a kind and what does not, would teach more than repeated descriptions of one object.

3. **Is the expectation tied to the noun?** We predicted that the index would be higher when the second object had the same name as the first than when it had a different name. A model can predict a kind's fixed shape from its noun alone, without comparing nouns. We therefore also predicted that models would tie the expectation to the noun when a kind's shape changed from one stretch of text to the next, and only the context showed which shape would come.

4. **What do natural input and pretrained models show?** We predicted that mentions of object kinds would be clustered in speech to young children, more so than in written text, and tested, two-sided, whether it differed from adult conversation. We predicted that in Pythia models, which are

trained on written text, the index would grow with size and over training, and that models trained on child-scale data would show it. We also predicted that larger Pythia models would tie the expectation to the noun.

We registered every confirmatory test before the models concerned were trained or scored, or before the transcripts were read. Some registrations were written after earlier results had been seen; Appendix A gives the dated record of what was known at each.

The paper makes three contributions:

1. With content held fixed, it shows that the arrangement of training text decides which feature small transformers expect to persist across objects. Random order, against our prediction, taught them to expect a shape to change; grouping by kind taught them to expect it to stay; and runs of different objects of a kind taught this more selectively than runs re-describing one object.
2. It shows when the models tie this expectation to the noun: little when each kind keeps one feature, and, for nouns seen in training, almost as much as the data warrant when only the context shows which feature will come.
3. It measures the same arrangement statistic in natural corpora, as a descriptive comparison. Models pretrained on written text show only a weak, partial analogue.

Our key finding is that arrangement alone decided which feature the models expected to persist: with the same sentences, the index was −1.56 nats in random order and 9.52 when the sentences about each kind were grouped. The question is therefore not whether a next-word learner acquires an expectation about kinds from text, but which arrangement teaches which expectation, and when the noun is needed to carry it. The models tied the expectation to the noun about ten times more strongly when a kind's shape changed between stretches of text (11.24 against 1.04 nats).

Our claims are about how transformers learn. We use the shape bias as a test, as Ritter et al. (2017) did for image classifiers, and we treat speech to children as one natural corpus among others, not as the thing being explained.

2. Background

2.1 The shape bias as a learned overhypothesis

In the original studies, children and adults extended a novel name to objects of the same shape more readily than to objects of the same size or texture (Landau, Smith, and Jones 1988). In toddlers, attention to shape grows as their vocabulary of object names grows (Gershkoff-Stowe and Smith 2004). Teaching 17-month-olds a few categories organised by shape made the bias appear early and sped up their learning of object names outside the laboratory (Smith et al. 2002). Teaching varied exemplars of each category helped more than teaching near-identical ones (Perry et al. 2010). The bias therefore looks learned, from the way named kinds vary.

Kemp, Perfors, and Tenenbaum (2007) made this precise. In their hierarchical Bayesian model each kind has its own distribution over features, and a higher level describes how variable kinds tend to be on each dimension. After a few kinds that are uniform in shape and varied in colour, the model expects a new kind to be uniform

in shape after one exemplar. Associative networks trained on the structure of early noun vocabularies acquire similar expectations (Colunga and Smith 2005), and neural networks trained on images (Ritter et al. 2017) or on artificial objects (Feinman and Lake 2018) can show a shape bias. These studies used perceptual input, or features given directly to the network. We ask whether a next-word learner acquires the regularity from descriptions in text, and what arrangement of the text it needs.

2.2 Artificial languages and novel words

Artificial languages let the training data be controlled exactly, so that a model's preferences can be traced to its input (McCoy, Frank, and Linzen 2020; White and Cotterell 2021; Kallini et al. 2024). These studies vary the grammar of the language; we keep the content fixed and vary only the order of the sentences. Our test is also a form of novel-word learning. Language models acquire words largely by frequency (Chang and Bergen 2022), can generalise a new verb from one or two sentences (Thrush, Wilcox, and Levy 2020) and can learn new words in context when trained to do so (Wang et al. 2025), but often fail to pass a property from a concept to its subtypes when the context contains distracting information (Misra, Rayz, and Ettinger 2023).

2.3 Training on speech to children

A line of work trains language models on data closer to a child's input. A small masked model trained on 5 million words of child-directed speech learned grammar about as well as a much larger model (Huebner et al. 2021). The BabyLM Challenge set limits of 10 and 100 million words (Warstadt et al. 2023; Hu et al. 2024). Models trained on child-directed speech capture its surface statistics but generalise questions linearly rather than hierarchically (Yedetore et al. 2023). Models trained on one child's speech and video learn word classes and word-object mappings (Wang et al. 2023; Vong et al. 2024).

Two findings bear on arrangement. Feng, Goodman, and Frank (2024) found that the local order of child-directed speech affected what models learned, while its global order by the child's age did not. Speech to children often comes in variation sets, runs of nearby utterances that repeat and vary talk about the same thing (Küntay and Slobin 1996). Adults learned an artificial language better when its input contained variation sets (Onnis, Waterfall, and Edelman 2008), but adding generated variation sets to the training data of small language models had mixed effects (Haga et al. 2024).

2.4 Burstiness, locality and in-context learning

Words are bursty: once a word appears, it is likely to appear again soon. Cache language models used this to predict speech (Kuhn and De Mori 1990), and models of word counts that allow it fit text better than those that do not (Church and Gale 1995; Madsen, Kauchak, and Elkan 2005). Church (2000) measured this as adaptation, the chance that a word recurs in a document once it has appeared. Our clustering measure, B (Section 3.1), is close in spirit. Language models show similar effects: with an added adaptation step, a recurrent model predicts words and structures better after recent exposure (van Schijndel and Linzen 2018), and transformers show structural priming (Sinclair et al. 2022).

In transformers, burstiness changes what is learned. Chan et al. (2022) trained transformers on sequences of images and found that in-context learning emerged only when items of a class appeared in bursts. Reddy (2024) traced this to the abrupt formation

of an induction head, and Singh et al. (2023) found that in-context learning can fade as learning stored in the weights takes over. Prystawski, Li, and Goodman (2023) showed that training on local samples from a structured distribution makes intermediate reasoning steps useful. In pretraining, putting related documents in the same window improves use of the context (Shi et al. 2024; Zhao et al. 2024). These studies concern what a model does with examples in its context, and they point to the value of local structure without implying a common mechanism. We ask whether training sets up an expectation about which feature persists before the context offers any evidence about it.

2.5 Order and curricula

Work on curricula asks whether the order of training examples matters, with mixed results. Starting small helped a recurrent network in one study (Elman 1993) but not under more natural conditions (Rohde and Plaut 1999); easy-to-hard ordering can speed learning (Bengio et al. 2009); and at child scale, curricula based on human notions of difficulty have given inconsistent gains (Diehl Martinez et al. 2023), while ordering by each example's estimated influence on the model helped (Schoenegger et al. 2025). Human category learning offers a parallel. Blocked study helps learners find what members of a category share, and interleaved study what tells categories apart (Carvalho and Goldstone 2014). For children, spaced presentations helped category learning (Vlach, Sandhofer, and Kornell 2008), but repeating a new word across successive sentences helped word learning (Schwab and Lew-Williams 2016).

3. Study 1: Small models trained on an artificial language

Study 1 asks what a transformer learns about kinds when its input contains nothing else, and how the arrangement of that input matters. We built corpora in which every regularity and the order of every sentence are under our control, and trained models small enough to train many times. Content is held fixed across orders of the same corpus and varied deliberately between corpora.

3.1 Corpora

Each corpus describes 32 kinds of object, each named by a novel noun (*wug*, *dax*, *blorp* and so on), with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a novel word from a list of ten per dimension, and is described by one sentence in one of five frames:

1. *A wug is a [shape] [colour] [texture] thing.*
2. *Look at the wug. It is [shape] and [colour].*
3. *Here is another wug. This one is [shape].*
4. *The wug I showed you is [shape], but a different [colour].*
5. *That is a wug because it is [shape].*

Shape is named in all five frames, colour in three and texture in one. The frames are the same in every corpus we compare, so they cannot by themselves explain a difference between corpora, although they may interact with what is manipulated.

The 448 sentences of a corpus, each used once, make up one pass. Five corpora were generated for each condition, one per seed.

Content. The corpora differ in which feature, if any, stays fixed within a kind (Table 1). In **Regular** corpora every kind keeps one shape across its 14 exemplars, while colour and texture vary. In **Mixed-slot** corpora each kind keeps one word from the shape list, but for about half the kinds that word appears in the texture slot, and colour varies; no single dimension is fixed across all kinds. In **No stable feature** corpora, built from Regular, each exemplar's shape is drawn at random from the list, so nothing is fixed within a kind. In **Colour stable** corpora, also built from Regular, shape is drawn in the same way and each kind keeps one colour instead. Frame 4 is unchanged, so in Colour stable corpora it names the kind's fixed colour after the words *a different*.

Objects. A further set of corpora separates two kinds of nearby repetition. In the Regular corpora, two sentences about a kind always describe different exemplars. In speech, nearby mentions often return to the same object. The Objects corpora keep the Regular content but give each kind 13 objects and describe each object four times, once in each of frames 1, 2, 3 and 5. Frame 4 is left out because it states that the colour differs. A pass of these corpora has 1,664 sentences.

Order. The same sentences were arranged in several orders. In **random** order every pass is shuffled. In **grouped** order, within each pass, the sentences about each kind are shuffled, cut into runs of four, and the runs are shuffled across kinds (Figure 1). Every sentence occurs exactly as often as in random order. In **graded** orders each run is kept with a set probability and otherwise split into single sentences before shuffling, which gives intermediate levels of clustering. In the Objects corpora, a **same-object run** is the four descriptions of one object, and a **different-object run** is four different objects of one kind, one per frame. A Latin square over objects and frames ensures that every sentence is used once per pass in every order. Same-object and different-object runs were each arranged fully grouped and at an intermediate level.

Reassigned. In every corpus so far, a kind's fixed feature is fixed for the whole corpus, so the noun alone predicts it, and a model can predict the shape in a sentence from its noun without comparing the sentence with the one before. In Reassigned corpora, built from Regular, each kind receives a new shape, drawn at random from the ten, in every pass, and keeps it in all of its sentences in that pass. The next sentence about the same kind still repeats the shape, but the noun no longer says which shape it will be. These corpora were arranged in random and grouped order.

Fresh and fixed text. A model trained many times over on one short file can memorise it. Our main models were therefore trained on fresh text: 7,500 passes of a corpus (2,100 for the Objects corpora), each arranged with its own random draw by the rule of its order, 34 to 35 million tokens in all. Training read this text in non-overlapping 128-token windows, so no window was seen twice. A second set of models was trained in the common way, on one fixed file of 11,648 sentences (26 passes, or 7 for the Objects corpora; about 120,000 tokens), with windows starting at random positions, so that each position was seen 160 to 290 times. These fixed-file models memorised their files. Their loss was 0.08 to 0.11 nats per token on their own file but 0.66 to 1.26 on a new arrangement of the same sentences, while fresh-text models scored 0.28 to 0.55 on both (Table C2). Table 1 shows which conditions were trained on fresh text and which on a fixed file; where both exist, we report fresh text first. The fresh graded order, built by the same rule as the fixed one at B = 3.6, measured B = 3.5.

Clustering. We measure clustering with one statistic, B, here and in Study 2. Take the sequence of mentions of object nouns in a text, and every pair of consecutive mentions that are no more than 100 words apart. P is the share of these pairs that name

Table 1
Study 1 corpora. Every corpus describes 32 kinds. Fresh: 7,500 passes (2,100 for Objects), each arranged anew and read once in non-overlapping 128-token windows. Fixed: one file of 11,648 sentences (10,304 to 12,992 for three further controls: No kind label, Paraphrased and Weak label), read in 128-token windows at random positions unless marked. Each order of each corpus was trained at three sizes with five seeds (15 models). The further controls are reported in Appendix D (Table D1).

Corpus	What stays fixed within a kind	Orders (measured B)	Text	Models
Regular	Shape. Colour and texture vary.	Random (0.9); graded (3.5); grouped (23)	Fresh	45
No stable feature	Nothing	Grouped (23)	Fresh	15
Objects	Shape; each object described four times	Same-object runs (24); different-object runs (24)	Fresh	30
Reassigned	Shape within a pass; a new shape for each kind in every pass	Random (0.9); grouped (23)	Fresh	30
Regular	Shape	Random (0.9); graded (1.4, 3.6, 7, 12); grouped (23)	Fixed	90
Regular	Shape	Random (0.9); graded (3.6)	Fixed, 512-token window	30
Mixed-slot	One shape-list word, in the shape slot for about half the kinds and the texture slot for the rest	Random; grouped	Fixed	30
No stable feature	Nothing	Random; grouped; random with a 512-token window	Fixed	45
Colour stable	Colour. Shape varies.	Random; grouped	Fixed	30
Objects	Shape; each object described four times	Random (1.0); same-object runs (3.6, 24); different-object runs (3.6, 24)	Fixed	75
Further controls: no kind label, weak label, paraphrased, frequency matched, added noise, feature swap	As Regular, with one change each	Random	Fixed	90
Total				510

the same kind. P_0 is the same share after the sentences (or utterances) of each document are shuffled, averaged over shuffles. B = P/P_0, pooled over documents. For example, in *Look at the dog. The dog is big. Where is the cup? Here is another dog.*, the consecutive pairs are dog–dog, dog–cup and cup–dog, so P = 1/3. B = 1 means no clustering beyond what the content predicts. Random-order Regular corpora have B = 0.9, slightly below 1, because each pass is shuffled separately: within a pass, a sentence's kind has 13 other members among 447 sentences, against 363 among 11,647 when the whole corpus is shuffled. The graded levels (Table 1) were chosen after the clustering of natural input had been measured (Study 2): 1.4 and 3.6 match B in written text (1.43) and in speech to children (3.57). They were fixed from the corpora alone before any graded-order model was trained.

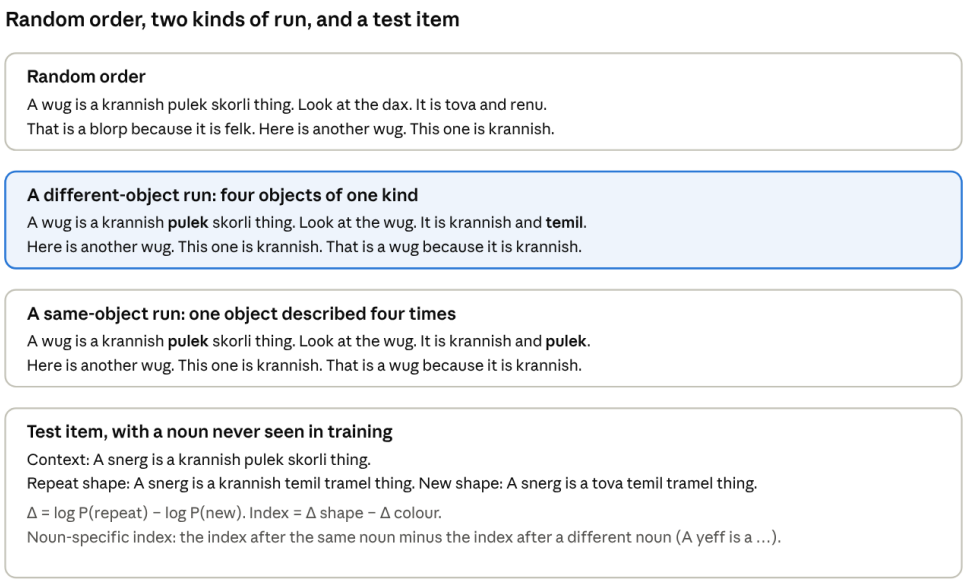


Figure 1
Illustrative sentences in the style of a random-order corpus, a different-object run and a same-object run, and a test item. In the different-object run the colour changes from one wug to the next; in the same-object run it repeats.

3.2 Models and training

We trained decoder-only transformers of three sizes: 3.4 million parameters (4 layers, 4 heads, width 256), 11.0 million (6 layers, 8 heads, width 384) and 25.6 million (8 layers, 8 heads, width 512). They were trained for 5,000, 8,000 and 10,000 steps with batches of 32, 32 and 16, peak learning rates of 3×10^{-4}, 2×10^{-4} and 1×10^{-4}, AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up, cosine decay and dropout of 0.1. Each training example is a 128-token window, which spans several sentences, so attention crosses sentence boundaries. The tokeniser for the Regular corpora has 351 entries (314 to 339 for the other corpus families), and every artificial word is a single token. A seed sets the corpus, the initial weights and the batch order (Table 1 gives the models per condition). For the window test, models were trained with 512-token windows and a quarter of the batch size, so that every step saw the same number of tokens. All models have separate input and output embeddings, and the parameter counts include both; the model code ties the two, but the tie was lost on the training hardware (Appendix A). Table C1 lists every setting.

3.3 Measuring the expectation

Eight novel nouns that never occur in training (*snerg*, *twick* and so on) are used in test items. They are in the vocabulary, so each is one token, but their input embeddings were never trained; weight decay shrinks them by less than 10% over training, so they remain distinct random vectors. With the intended tie, the same vectors would have

served as the novel nouns' output weights and would have changed in training; we cannot say whether the models would then have tracked these nouns. Novel nouns keep learned knowledge about a noun out of the test, but the test then depends on the model registering that the same untrained token recurs. We therefore also test with nouns from training, whose embeddings are trained (Appendix F). Each item gives the model one sentence in frame 1 describing an exemplar, followed by two continuations that differ in one word:

> Context: *A snerg is a krannish pulek skorli thing.* Repeat: *A snerg is a* ***krannish*** *temil tramel thing.* New: *A snerg is a* ***tova*** *temil tramel thing.*

In this item the probed dimension is shape. The repeat continuation keeps the context's shape, the new one does not, and both give new values for the other two dimensions. The model scores each continuation by the sum of the log probabilities of its tokens given the context, so the difference, Δ, includes the changed word and its effect on the words after it. Δ is the model's preference for repeating the probed dimension, in nats. Trained models generally prefer words they have just seen, so Δ alone mixes the dimension with copying. Our index is the difference between dimensions:

$$\text{index} = \Delta_{\text{shape}} - \Delta_{\text{colour}}$$

Copying that affects shape and colour equally cancels. Copying that differs between dimensions does not, which is one reason every conclusion rests on comparisons between corpora tested in the same way, and why we report Δ for each dimension as well. The shape-over-texture index is defined in the same way. With one exemplar, nothing in the context shows which dimension is fixed, so differences between trained conditions come from training.

Every item is also given with a different novel noun in the continuations (*A yeff is a krannish temil tramel thing*). The **noun-specific index** is the index after the same noun minus the index after a different noun. A kind-tied expectation scores above zero on it; a noun-blind one scores zero.

With 25 draws of features per noun, three probed dimensions and two continuation nouns there are 1,200 one-exemplar items. A further 2,400 items with two or four exemplars test the use of evidence in the context (Table E2). Untrained models of the same architecture, with the Regular tokeniser, give an index of 0.02 nats (SD 0.07).

3.4 Statistics

Each model gives one index, its mean over items. Models of the three sizes trained with the same seed share a corpus, so the five seeds are the units of replication. We average the three sizes within each seed and compare conditions by paired t tests over the five seed means (4 degrees of freedom); the differences and 95% intervals given in the text are these seed-level estimates. Registered contrasts were tested one-sided, with Holm (1979) correction within each registered family. For fresh-text models the registered test was this seed-level test. For fixed-file models it was a linear mixed model (index $\sim$ condition + size, with a random intercept for seed) or, for some contrasts, a Welch t test over models; Table E1 gives the seed-level results beside them. Single conditions were compared with zero by one-sample t tests. Results reported with a Holm-corrected p, or called registered, were registered; all others are exploratory. We registered 0.2 nats

Table 2
Study 1 results for models trained on fresh text, in nats after one exemplar with the same noun; 15 models per row. Δ is the preference for repeating a dimension. Table E4 gives the same for every fixed-file condition.

Corpus	Order (B)	Δ shape	Δ colour	Shape over colour, mean (SD)	Models above zero	Shape over texture
Regular	Random (0.9)	−1.55	0.01	−1.56 (1.19)	0 of 15	−1.44
Regular	Graded (3.5)	1.18	−0.10	1.28 (1.83)	13	1.21
Regular	Grouped (23)	6.88	−2.64	9.52 (3.06)	15	6.56
No stable feature	Grouped (23)	−3.88	0.10	−3.98 (2.52)	1	−4.32
Objects	Same-object runs (24)	6.87	0.99	5.87 (2.46)	15	6.73
Objects	Different-object runs (24)	8.05	0.46	7.60 (2.42)	15	7.91
Reassigned	Random (0.9)	0.22	0.02	0.21 (0.49)	10	0.39
Reassigned	Grouped (23)	6.81	−2.99	9.80 (3.97)	15	5.62

as the smallest effect of interest and tested equivalence to zero against it with two one-sided tests (Lakens 2017). This bound is small next to the effects reported here, so the equivalence tests are strict.

3.5 Results

Table 2 gives Δ for shape and colour and both indices for every fresh-text condition, and Table E4 for every fixed-file condition. Every contrast reported below as reliable had the same direction at all three model sizes (Table E2), and we report sizes pooled.

Arrangement decides which feature persists. In random order, Regular models came to expect the shape to change: their shape-over-colour index was −1.56 nats [−2.45, −0.67], below zero in all 15 models, against our prediction of an index above zero. The same sentences in grouped order gave 9.52 nats, higher in all five seeds and at every model size (difference 11.08 [9.57, 12.59], Holm $p < .001$). Clustering close to the level of object labels in speech to children (B = 3.5 here; 3.57 in Study 2) already gave 1.28, above random order (difference 2.84 [1.41, 4.27], Holm $p = .005$). Fixed-file models showed the grouping effect at about half the size (6.49 against 0.47; difference 6.02 [5.24, 6.79], Holm $p < .001$), and their random-order index was not reliably above zero (Holm $p = .061$). Early in their training, while their loss was near that of the fresh-text models, their random-order index was −0.4 to −1.5 nats; it rose above zero only late in training, when their loss fell to the memorised level (exploratory).

The gain depended on a feature that was fixed within kinds. When no feature was fixed, grouped models expected the shape to change, at −3.98 (difference from grouped Regular 13.50 [11.48, 15.52], Holm $p < .001$). Fixed-file models, the only ones trained on Colour stable corpora, show that the gain follows whichever feature is fixed. Grouped Colour stable models preferred colour, at −3.20, below grouped No stable feature models (difference 2.76 [1.99, 3.52], Holm $p < .001$), and none of the 15 was above zero. Their components show why: they expected colour to repeat (Δ 2.39) and shape not to (−0.81).

Mixed-slot corpora, trained as fixed files, keep a fixed word for every kind but no single fixed dimension. In grouped order these models preferred shape over colour (2.12), as their corpus supports, since colour is never fixed there, but showed no reliable preference between shape and texture (0.46), each of which is fixed for about half the



Figure 2
Shape-over-colour index (nats, one exemplar, same noun) against clustering B in the Regular corpora. Grey points and solid blue line: fixed-file models in six orders (points, mean of the five models at each size; line, mean of all 15). Open circles and dashed line: fresh-text models in three orders (mean of 15). Vertical lines and shaded bands: effective clustering of different objects in the three natural corpora (Study 2), with 95% intervals, placed as registered. The overlay places the natural values on the same axis; it does not predict what a model trained on them would learn.

kinds. Grouped Regular models were above them on both indices (differences 4.37 [1.98, 6.76] and 4.89 [2.69, 7.10], Holm $p < .001$). In random order, Regular models were above Mixed-slot models by the registered test (0.47 against −0.25, Holm $p = .032$), but not reliably at the seed level (3 of 5 seeds; $p = .11$).

Across the six orders of the fixed-file Regular corpora, the index rose with clustering at every step (Table E4, Figure 2). The registered test, a linear trend in the natural log of B, was supported (slope 1.63 nats per log unit, SE 0.10, Holm $p < .001$; seed level, 1.63 [1.46, 1.80]). The rise was steeper at high clustering: about 0.6 to 1.0 nats per log unit up to B = 7, and 2.7 and 4.6 nats per log unit over the two highest steps. At B = 3.6, fixed-file models were above random order (1.43 against 0.47; difference 0.96 [0.63, 1.28], Holm $p < .001$), as were fixed-file Objects models with different-object runs at the same level (1.18 against 0.22; difference 0.96 [0.37, 1.54], Holm $p < .001$). Fresh-text models agree with this curve at B = 3.5, but lie below it in random order and above it in grouped order (Figure 2).

The corpora show what the models had to go on. For every pair of consecutive sentences we counted how often the second repeats the first one's shape or colour (exploratory; Table E3). In grouped order the next sentence is about the same kind 72% of the time, and then always repeats the shape. In random order it is about the same kind

3% of the time, and a sentence about another kind repeats the previous shape slightly less often than chance (7%, against 10% for colour). A novel test noun gives the model no way to tell whether the second object is of the same kind, so it can only apply what the next sentence usually does. A learner that predicted each feature from the previous sentence alone, ignoring the noun, would show an index from −3.26 (grouped Colour stable) to 3.43 (grouped different-object runs). The model index tracked this reference across conditions (Pearson r = .88 over the 8 fresh-text conditions and .95 over the 12 fixed-file conditions; exploratory and descriptive, since conditions share seeds, and driven mainly by the contrast between grouped and ungrouped orders). The model index sums over the whole continuation and the reference concerns one feature word, so only their agreement across conditions is comparable. The reference does not explain the negative indices. It is about zero for fresh random order (−0.01) and for grouped No stable feature corpora (0.02), where the models gave −1.56 and −3.98, and we have no tested account of these values. Grouped same-object runs also departed from it: the models repeated shape, which every frame names, more than the corpus warrants.

The models were also sensitive to how long runs last. When two or four sentences about the novel kind, sharing a shape, preceded the test sentence, the index of grouped Regular models fell, from 9.52 to 7.88 and 4.24 nats (Table E2), although a learner of the overhypothesis should grow more confident with more evidence. In grouped order the sentences about a kind come in runs of four: after one sentence about a kind the next is about the same kind in every case, after two in 76% of cases, and after four in 3% (exploratory counts on the fresh text). This fits models that have learned that runs end after about four sentences.

What kind of grouping helps. In fixed-file models, under our matched-tokens recipe, a longer window did not raise the index (with tokens per step held fixed, the 512-token models also saw a quarter as many sequences per step, so the result applies to this recipe, not to window length alone; Table C1). With 512-token windows, which hold about four times as many sentences, random-order Regular models gave 0.50 nats against 0.47 with 128 tokens (difference 0.03 [−0.44, 0.50], Holm p = .87), and the registered equivalence test did not show equivalence either (p = .15). At B = 3.6, the longer window gave less, 0.80 against 1.43 (difference −0.63 [−0.90, −0.37]; two-sided seed-level p = .003). With the longer window, Regular models were still above No stable feature models (0.50 against 0.10, Holm p = .045; seed level, 4 of 5 seeds, p = .032).

At matched clustering, runs of different objects gave a larger index than runs re-describing one object: 7.60 against 5.87 nats in grouped order (difference 1.72 [0.54, 2.90], Holm p = .008; higher in all five seeds). Fixed-file models showed the same (5.65 against 2.99; difference 2.66 [1.57, 3.75], Holm $p < .001$), and also at B = 3.6 (1.18 against 0.45; difference 0.73 [0.15, 1.32], Holm $p < .001$). Their same-object runs at B = 3.6 were neither reliably different from random order (0.45 against 0.22; difference 0.22 [−0.30, 0.74]) nor equivalent to it (registered secondary test).

The components show how the two kinds of run differ (Figure E1). After same-object runs, models expected colour to repeat more than after different-object runs (Δ colour 0.99 against 0.46 with fresh text; 1.65 against −0.28 with fixed files), because in those runs every feature does repeat. Their index stays positive because they repeated shape, named in all four object frames, more than colour, named in two. Part of this may be copying.

Is the expectation tied to the noun? With novel nouns, the index was about the same after the same noun and after a different noun in every corpus with a fixed shape.

In grouped Regular models the noun-specific index was 0.12 nats [0.04, 0.21], against an index of 9.52. In fixed-file random-order models it was 0.005, equivalent to zero (registered test; p = .001 for equivalence within ± 0.2). This does not show that the expectation ignores the noun. The input embeddings of the novel nouns were never trained, and the models showed little sign of telling these nouns apart: none of the 60 fixed-file models we checked predicted a novel noun again after reading it (exploratory; Appendix E).

Nouns from training have trained embeddings, but in corpora where each kind keeps one shape, the items give a trained noun a shape it never has in training, because each noun's own shape is left out of the items (Appendix F). On these items, grouped Regular models gave a noun-specific index of 1.04 [0.76, 1.31] (index 2.00 after the same noun against 0.97 after a different one). Most of it came from colour: after the same noun they expected colour to change more (Δ colour -1.94 against -1.04), while their expectation that shape would repeat barely differed (Δ shape 0.06 against -0.08). In these corpora each kind keeps one shape for the whole corpus, so the noun alone predicts the shape in a sentence about a trained kind, and a model has no need to compare the shape with the sentence before. Two conditions gave negative noun-specific indices on trained nouns (-2.60 in fresh grouped No stable feature corpora and -1.06 in fixed-file grouped same-object runs; Table F1), which we do not interpret.

Reassigned corpora remove that route: a kind's shape changes from pass to pass, so only the previous sentence about the kind shows which shape comes next. Trained on them in grouped order, the models tied the repetition of shape to the noun. With nouns from training, the noun-specific index was 11.24 nats [10.44, 12.03] (Holm $p < .001$), close to the 11.4 that the corpus warrants and about ten times the value in grouped Regular models. The models expected shape to repeat after the same noun (Δ shape 10.01) and not after a different one (0.08). With novel nouns the effect was small but reliable (0.37 [0.24, 0.49], Holm p = .001) and larger than in grouped Regular models (difference 0.25 [0.10, 0.39], Holm p = .005), though not in the largest models (difference 0.02). The noun-blind index was the same with and without reassignment (9.80 against 9.52). In random order, Reassigned corpora taught neither a preference for shape (0.21 [-0.31, 0.72]) nor a dependence on the noun (-0.07 with nouns from training), so the effect needed sentences about a kind to be close together.

Summary. Arrangement taught the models which feature persists: their index tracked what the next sentence usually does in each arrangement, although the negative indices go beyond it. The models tied this expectation to the noun when the noun was needed to predict the feature, and hardly at all when each kind kept one shape throughout.

4. Study 2: How clustered is natural input?

Study 2 measures, with the statistic B of Study 1, how clustered mentions of object kinds are in three kinds of natural input: speech recorded around young children, conversation between adults, and the written text language models are trained on. It is descriptive. B is a shared statistic, not a calibrated dose: natural corpora differ from the Study 1 corpora in many ways besides clustering.

4.1 Method

Corpora. We used the North American English collection of CHILDES (MacWhinney 2000), downloaded in full from TalkBank, kept every transcript whose target child was

under 36 months, and took every utterance not spoken by that child. Most of this speech is addressed to the child, but some is addressed to others present; for brevity we call it speech to children. For adult conversation we used CABank: the Santa Barbara Corpus of Spoken American English (Du Bois and Englebretson 2004), CallHome English (Canavan, Graff, and Zipperlen 1997) and CallFriend English (Canavan and Zipperlen 1996a,b). All but the Santa Barbara Corpus are telephone calls, in which the speakers share no objects in view. For written text we used the first 10,000 documents of the Pile (Gao et al. 2020), the training data of the Pythia models in Study 3, split into sentences. Documents of fewer than 400 words or with no mention of a target noun were excluded, leaving 2,116 transcripts from 35 CHILDES corpora, 245 CABank transcripts from four sources and 2,474 Pile documents from 21 subsets.

Kinds. We counted mentions of 70 count nouns from the MacArthur-Bates Communicative Development Inventory (Fenson et al. 2007) that name solid, inanimate objects, such as *ball*, *cup*, *shoe* and *truck*, the kinds for which children show the shape bias. Singular and plural count as one kind, and words often used as verbs or adjectives in speech to children, such as *brush* and *orange*, were left out. Because the nouns come from a list of words young children know, they are more frequent in speech to children than in the other corpora, by design.

Measure. B was computed as in Study 1, with the utterances or sentences of each document shuffled for P_0, within a window of 100 words, about the length of the 128-token training window of Study 1. Intervals resample sources (CHILDES corpora, CABank sources or Pile subsets) and then documents within them.

What repeated labels refer to. B counts repeated labels. It cannot tell two balls from one ball named twice, and Study 1 showed that the difference matters. The author coded 500 random same-label pairs (300 from speech to children, 100 from each of the others), with source labels withheld, as referring to the same object, to different objects of the kind, to the kind in general, or unclear; source could sometimes be guessed from the text. As a model-assisted consistency check, an instance of Claude (Anthropic), given only the coding rules and the pairs, coded a random quarter of them without access to the first coder's codes; the prompt and the codes are archived. Agreement on 124 pairs was 84% (Cohen's $\kappa = 0.78$). This is not a measure of agreement between human coders.

Effective clustering (exploratory). As a rough projection onto the Study 1 scale, we discounted each corpus's excess clustering by the share of coded pairs that referred to different objects, d, out of those that referred to particular objects: $B_{\text{eff}} = 1 + (\text{B} - 1)\text{d}$. A strict version takes d out of all coded pairs. Intervals resample sources and documents for B and coded pairs for d. B_{eff} assumes that only different-object pairs inform the learner. Study 1 shows that same-object repetition is not neutral, so B_{eff} is a heuristic, not a corrected estimate.

4.2 Results

Object labels were clustered in every corpus, most of all in speech to children (Table 3). In speech to children B was well above 1 (registered test, Holm $p < .001$), above written text (difference 2.14 [1.55, 2.91], Holm $p < .001$) and above adult conversation (difference 1.24 [0.51, 2.03]; registered two-sided test, $p < .001$). The order of the three held in three registered secondary analyses: when each transcript was weighted equally, when every common noun was counted instead of our 70, and when documents were shuffled only within 500-word blocks, which keeps changes of topic. That last check lowered B in speech to children from 3.57 to 2.08, so part of the raw clustering reflects topic.

Table 3
Clustering of mentions of 70 object nouns within 100 words. Intervals resample sources and then documents, and for B_{eff} also coded pairs. B_{eff} is exploratory.

Measure	Speech to children	Adult conversation	Written text
B [95% CI]	3.57 [3.13, 4.30]	2.33 [2.02, 2.93]	1.43 [1.22, 1.86]
B, each transcript weighted equally	2.91	2.27	1.31
B, shuffling only within 500-word blocks	2.08	1.28	1.16
B, all common nouns	9.65	5.73	2.07
Same-label pairs per 1,000 words	10.1	0.9	1.0
Coded pairs: different objects	27% (82 of 300)	25% (25 of 100)	13% (13 of 100)
Coded pairs: same object	46% (139)	25% (25)	12% (12)
Coded pairs: kind in general or unclear	26%	50%	75%
B_{eff} [95% CI]	1.95 [1.73, 2.28]	1.66 [1.44, 2.04]	1.22 [1.09, 1.47]
B_{eff}, strict	1.70 [1.53, 1.96]	1.33 [1.21, 1.54]	1.06 [1.02, 1.13]

Nearly half of the repeated labels in speech to children referred again to the same object, and about a quarter brought a different object, a share similar to adult conversation. Most repeated labels in written text did not refer to a physical object, or referred to the kind in general. Effective clustering of different objects was therefore lower than B in every corpus. Speech to children was above written text on this measure (difference 0.73 [0.41, 1.09]) but not reliably above adult conversation (0.29 [−0.14, 0.69]).

Placed on the Study 1 scale by effective clustering, as registered, written text lies between random order and the graded order at B = 1.4, and adult conversation and speech to children lie between the graded orders at B = 1.4 and B = 3.6 (Figure 2). Over that range the index of fixed-file models rose only from 0.47 to 1.43 nats; fresh-text models were trained only near its ends, where they gave −1.56 in random order and 1.28 at B = 3.5. This does not say what a model trained on each corpus would learn.

The corpora differ more in how often such repetitions occur. Speech to children contained 10.1 same-label pairs per 1,000 words, against about 1 in the other two. Combining these rates with the coded shares, it contained about 2.8 nearby repetitions per 1,000 words that bring a different object of a kind, against about 0.2 in adult conversation and 0.1 in written text (exploratory). Part of this difference comes from our choice of nouns. B is a ratio and does not capture density, which Study 1 held fixed.

5. Study 3: Pretrained models

Study 3 asks whether models trained on large written corpora show the expectation, how it depends on size and training, and whether they tie it to the noun. It is a descriptive comparison: the arrangement of their training data is not manipulated.

5.1 Method

Items. The items follow the format of Study 1 in English. Each feature gets its own sentence, so English adjective order cannot favour one dimension, and the order of the three sentences is counterbalanced:

Context: *Here is a snerg. It is cylindrical. It is purple. It is smooth.* Repeat: *Here is another snerg. It is* ***cylindrical****. It is turquoise. It is fluffy.* New: *Here is another snerg. It is* ***circular****. It is turquoise. It is fluffy.*

Each dimension uses eight English words of similar frequency, and the nouns are novel words spelled with ordinary subword pieces (Appendix G). In the original items the same-noun continuation began *Here is another snerg* and the different-noun continuation *Here is a zull*, so the noun-specific index compared the noun and the word *another* together. Four further item sets with 16 nouns keep the wording of the two continuations the same: *a* in both (the registered primary set), *another* in both, a second frame, and a second word list.

Models. The Pythia models (Biderman et al. 2023; van der Wal et al. 2025) range from 14 million to 12 billion parameters and were trained on the same 300 billion tokens of the Pile in the same order. We scored their final checkpoints and checkpoints across training, and nine further training seeds of the five smallest sizes (van der Wal et al. 2025). We also scored the two GPT-2 baselines of the BabyLM Challenge (124 million parameters, trained on 10 and 100 million words; Charpentier et al. 2025), and Qwen2.5 72B (Qwen Team 2024) and Llama 3.1 70B (Grattafiori et al. 2024) as descriptive reference points. Appendix G lists every checkpoint.

Tests. Trends with size are Spearman correlations across Pythia sizes; trends over training are tested by permuting checkpoints within size; intervals resample nouns. The pooled noun-specific test uses a mixed model with crossed random effects for nouns, items and feature words.

5.2 Results

The index rose with size but not steadily over training. At the end of training the shape-over-colour index rose with Pythia size (Spearman $\rho = .92$, Holm p = .003), from −2.44 nats at 14 million parameters to 1.17 at 12 billion (Figure 3). This raw index includes effects of the particular words. Untrained Pythia models give about 0.5 nats, because in some colour pairs the repeated and the new word take different numbers of tokens, and an untrained model pays roughly the same cost for every token. On the original items whose two words take the same number of tokens (all shape pairs and a quarter of the colour pairs; exploratory), untrained models gave −0.32 to 0.31 nats, the trend with size held ($\rho = .93$), and every Pythia model from 1B upwards gave an index above zero, from 1.00 [0.04, 1.82] at 1B to 1.62 [0.86, 2.19] at 12B. The trend with size held on every item set, including the second word list ($\rho = .83$ to .95; exploratory). The index did not rise steadily over training (mean within-size $\rho = -.34$). At every size, training first lowered it, to about −2 nats between steps 1,000 and 4,000, much as in the random-order models of Study 1, and only the larger models recovered. The BabyLM baselines did not show a reliable index (10 million words, 0.47 [−1.05, 1.93]; 100 million words, 0.37 [−0.04, 0.73]; Holm p = .52 and .12). On the items whose two words take the same number of tokens (exploratory), the 100-million-word baseline gave 1.16 [0.52, 1.82]. Pythia 160M, of similar size, gave a negative index (−1.22 [−1.66, −0.75]) even after 300 billion tokens, but the two families differ in more than their data, so this comparison does not separate data from size.

Weak, wording-dependent evidence of noun binding. On the original items the noun-specific index was positive in every Pythia model from 410 million parameters, but those items confounded the noun with *another*. With *a* in both continuations, the registered tests were not supported: the index did not rise with size ($\rho = -.42$, Holm p

= 1.00), and pooled over Pythia 410M to 12B it was not above zero (−0.15 nats, SE 0.12, Holm p = 1.00). It was higher at 410M than at 160M across ten training seeds of each (0.15 against −0.15 nats, Holm p = .012), with 7 of the 10 seeds at 410M positive. With *another* in both continuations (a registered secondary analysis), it was above zero in four of the six Pythia models from 410M (0.28, 0.77, 0.30 and 0.17 nats at 410M, 1B, 2.8B and 6.9B), but not at 1.4B or 12B, and it was small and mixed in sign on the second frame and the second word list. Of the two reference models of about 70 billion parameters, Qwen2.5 gave a positive index with *a* in both continuations (0.56 [0.28, 0.84]) and Llama 3.1 did not (−0.16 [−0.36, 0.06]).

The components show why the wording matters. After *another*, which marks a new member of the kind, the larger models expected shape to repeat and colour to change more after the same noun than after a different one, and *another* also raised the raw index, to 0.49 to 1.08 nats in Pythia 1B to 12B (−0.34 to 0.46 with a in both). After *a*, the same noun may name the same object again, and then every feature should repeat: colour repeated more after the same noun, and the noun-specific index fell.

Texture and context. No Pythia, Qwen or Llama model preferred shape over texture, so what the large models show is a shape-over-colour expectation, not a general shape bias (image classifiers trained on ImageNet likewise favour texture over shape; Geirhos et al. 2019). The BabyLM baseline trained on 10 million words did prefer shape over texture (1.17 [0.24, 2.34]). The two largest Pythia models tied texture to the noun more than shape. When two or four exemplars in the context shared a shape, every trained model followed the context, strongly so from 410 million parameters. What developed with size was the expectation after one exemplar, not the use of evidence in the context.

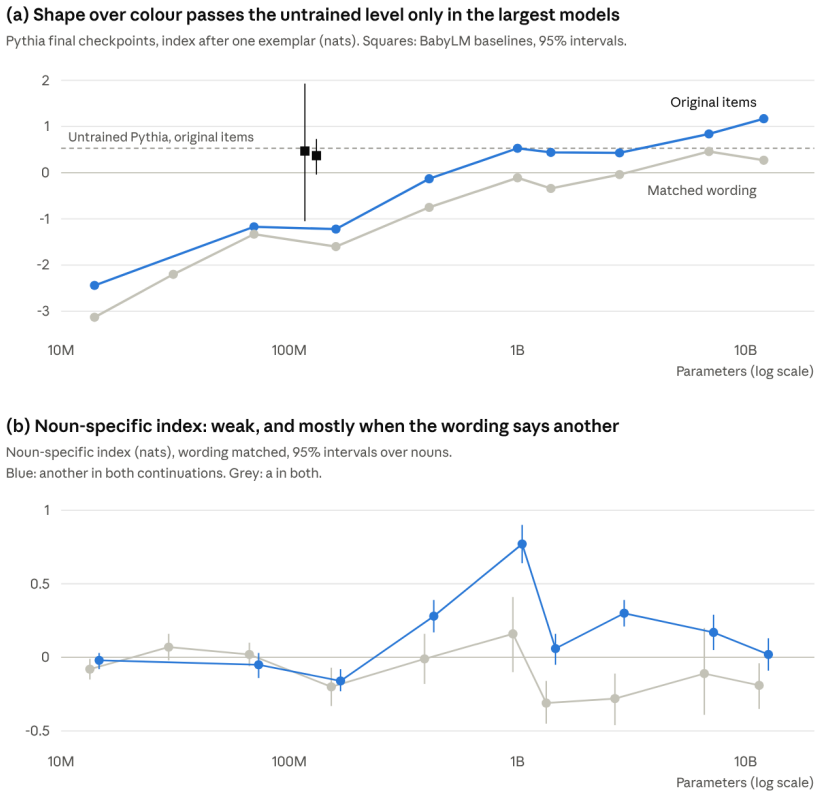


Figure 3
Pretrained models. (a) Shape-over-colour index (nats) at the end of training by Pythia size, on the original items and with matched wording (*a* in both continuations). Squares: BabyLM baselines with 95% intervals over nouns. Dashed line: untrained Pythia models on the original items (about 0.5 nats, from differences in token counts; Section 5.2). (b) Noun-specific index with matched wording, with *a* or *another* in both continuations, with 95% intervals over nouns. In both panels, the 31M model was scored only with *a* in both continuations.

6. Discussion

Table 4 summarises the main results.

What was learned. Small transformers learned what the next sentence usually does. In grouped order the next sentence is often about another member of the same kind; its fixed feature can be predicted from the sentence before and its varying features cannot, so the models learned to carry the fixed feature forward. They also became sensitive to how long runs last: with four sentences about a kind in the context, where a fifth is rare, the index fell from 9.52 to 4.24 nats. In random order the next sentence is nearly always about another kind, and the models expected its shape to differ, more strongly than this statistic alone predicts. Which feature was carried forward followed the content: the shape in Regular corpora, the colour in Colour stable corpora (fixed-file models), and none in corpora with no fixed feature, where the models instead expected the shape to change (−3.98 nats with fresh text). Our experiments identify the training conditions that produce the behaviour, not the mechanism inside the models. Carrying a word

Table 4
Main results. Study 1 estimates are seed-level differences, with the three sizes averaged within each of five seeds (t-based 95% intervals, one-sided tests, 4 degrees of freedom), for fresh-text models unless marked; for the clustering row, the mean of the five per-seed slopes. The registered p is the test named in the registration, Holm-corrected within it; Secondary marks a registered secondary analysis. Corpus intervals resample sources and then documents. Appendix A, Table A4, gives all 32 registered hypotheses.

Result	Estimate [95% CI]	Seed-level p	Registered p
Grouped order above random order	11.08 [9.57, 12.59] nats	$< .001$	$< .001$
Clustering at B = 3.5 above random order	2.84 [1.41, 4.27]	.003	.005
Grouped: Regular above No stable feature	13.50 [11.48, 15.52]	$< .001$	$< .001$
Grouped: Colour stable below No stable feature (fixed files; preference reverses to colour)	2.76 [1.99, 3.52]	$< .001$	$< .001$
Index rises with clustering (fixed files; slope per unit of log B)	1.63 [1.46, 1.80]	$< .001$	$< .001$
Different-object above same-object runs, grouped	1.72 [0.54, 2.90]	.008	.008
Same noun above different noun, shape reassigned, nouns from training	11.24 [10.44, 12.03]	$< .001$	$< .001$
The same, novel nouns	0.37 [0.24, 0.49]	$< .001$	.001
Same noun above different noun, shape fixed (grouped Regular), nouns from training	1.04 [0.76, 1.31]	$< .001$ (two-sided)	Secondary
Speech to children above written text, B	2.14 [1.55, 2.91]		$< .001$
Pythia: shape over colour rises with size	$\rho = .92$		.003

forward from the previous sentence is the kind of behaviour attributed to induction heads (Olsson et al. 2022), but we did not examine attention.

The noun. Whether the models tied this expectation to the noun depended on whether the data required it. When each kind kept one shape for the whole corpus, the noun alone could predict a trained kind's shape, and the models barely tied shape repetition to the noun. When a kind's shape changed between passes, only the previous sentence about the same kind showed which shape would come, and the models tied the repetition of shape to the noun almost exactly as much as the text warranted. This is the expectation an overhypothesis about named kinds requires: an object with the same name will share a shape that the learner has seen only once. It appeared here for nouns with trained embeddings; with untrained test nouns it was small. A similar trade-off between what is stored in the weights and what is read from the context appears in transformers trained on sequences of images. There, in-context learning emerged when items of a class appeared in bursts among many rare classes, and more strongly when the meanings of items were dynamic rather than fixed, and it faded as learning in the weights took over (Chan et al. 2022; Singh et al. 2023). Our Reassigned corpora are an analogue of dynamic meaning for the features of a kind, with the content of the text otherwise held fixed.

Fixed and fresh text. Small models trained on one fixed file memorised it, and their effects differed in size and, in random order, in sign. Experiments that train small models on a short corpus for many passes may therefore measure the memorised

sequence as well as the statistics of the language. Arranging every pass afresh separates the two at no cost in content.

Same-object repetition. When the next sentence describes the same object, every feature repeats, and the right prediction is to copy all of them. Same-object runs taught the models to expect colour to repeat as well, although they repeated shape more than the corpus warrants. Repetition teaches selectively which feature persists when it brings different objects of a kind; re-describing one object teaches that colour persists too, and so teaches less selectively. How often a kind recurs is not enough; what recurs matters.

Natural input. Speech to children was more clustered than written text, but nearly half of its nearby repetitions returned to the same object, so its effective clustering of different objects was only modestly higher. It differed more in density, partly because of our choice of nouns. Study 1 held density fixed, so we do not know how much density adds.

Pretrained models. Pythia models gained a shape-over-colour expectation with size, but none preferred shape over texture, and evidence that they tied it to the noun was weak and depended on wording. We did not arrange their training data, so this is a comparison, not a test.

Training data. Arrangement can be changed without changing content. Beyond packing related documents into one window (Shi et al. 2024; Zhao et al. 2024) or ordering by difficulty (Diehl Martinez et al. 2023), our results suggest a specific aim for regularities that hold across items: place different instances of a kind near each other, rather than repeated descriptions of one instance. They also suggest when a learner will tie such a regularity to a name. If each named kind can be learned one by one, it need not; if what a name predicts changes from one context to the next, as for many new kinds each seen only briefly, it must (compare meta-learning approaches to word learning; Wang et al. 2025).

7. Limitations

The artificial language of Study 1 has 32 kinds, three dimensions, one fixed feature shared by every kind and five sentence frames. The frames name shape more often than colour or texture, and frame 4 states that a colour differs, so conclusions rest on comparisons between corpora that share the frames, and Δ for one dimension is not a model's intrinsic preference among dimensions; counterbalancing the frames would remove this asymmetry. The models are small and were trained with one recipe, with five seeds per condition. Several conditions were trained only on fixed files (Table 1), which their models memorised; where both exist, contrasts have the same direction with fresh text and fixed files, but their sizes differ, and the random-order index differs in sign. The novel test nouns had untrained input embeddings, because a tie between the input and output embeddings was lost on the training hardware, so the noun results rest on nouns from training. That the Regular models predicted each trained kind's shape from its noun is an untested inference. Frame 3 says *another* even when a same-object run re-describes the same object. B counts repeated labels, not referents, and effective clustering rests on 500 pairs coded by one person, with a model as second coder. Three of the four adult sources are telephone calls, and all corpora are English. Study 3 does not manipulate the arrangement of pretraining data, so it cannot show that clustering causes the pattern it reports.

Acknowledgments

Generative AI disclosure: Claude (Anthropic) was used to help design analyses, write code, draft and edit text and check citations, and as the second coder in Study 2. The author is responsible for all content, including the accuracy of every citation.

References

Bengio, Y., J. Louradour, R. Collobert, and J. Weston. 2009. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*, pages 41–48. https://doi.org/10.1145/1553374.1553380.

Biderman, S., H. Schoelkopf, Q. G. Anthony, H. Bradley, K. O'Brien, E. Hallahan, M. A. Khan, S. Purohit, U. S. Prashanth, E. Raff, A. Skowron, L. Sutawika, and O. van der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of Proceedings of Machine Learning Research, pages 2397–2430.

Canavan, A., D. Graff, and G. Zipperlen. 1997. CALLHOME American English Speech (LDC97S42). Linguistic Data Consortium. https://doi.org/10.35111/exq3-x930.

Canavan, A. and G. Zipperlen. 1996a. CALLFRIEND American English-Non-Southern Dialect (LDC96S46). Linguistic Data Consortium. https://doi.org/10.35111/d37s-c536.

Canavan, A. and G. Zipperlen. 1996b. CALLFRIEND American English-Southern Dialect (LDC96S47). Linguistic Data Consortium. https://doi.org/10.35111/a8vf-6k58.

Carvalho, P. F. and R. L. Goldstone. 2014. Putting category learning in order: Category structure and temporal arrangement affect the benefit of interleaved over blocked study. *Memory & Cognition*, 42(3):481–495. https://doi.org/10.3758/s13421-013-0371-0.

Chan, S. C. Y., A. Santoro, A. K. Lampinen, J. X. Wang, A. K. Singh, P. H. Richemond, J. L. McClelland, and F. Hill. 2022. Data distributional properties drive emergent in-context learning in transformers. In *Advances in Neural Information Processing Systems 35*, pages 18878–18891. https://doi.org/10.52202/068431-1371.

Chang, T. A. and B. K. Bergen. 2022. Word acquisition in neural language models. *Transactions of the Association for Computational Linguistics*, 10:1–16. https://doi.org/10.1162/tacl_a_00444.

Charpentier, L., L. Choshen, R. Cotterell, M. O. Gul, M. Y. Hu, J. Jumelet, T. Linzen, J. Liu, A. Mueller, C. Ross, R. S. Shah, A. Warstadt, E. G. Wilcox, and A. Williams. 2025. BabyLM turns 3: Call for papers for the 2025 BabyLM workshop. ArXiv:2502.10645.

Church, K. W. 2000. Empirical estimates of adaptation: The chance of two Noriegas is closer to $p/2$ than p^2. In *COLING 2000 Volume 1: The 18th International Conference on Computational Linguistics*. https://aclanthology.org/C00-1027.

Church, K. W. and W. A. Gale. 1995. Poisson mixtures. *Natural Language Engineering*, 1(2):163–190. https://doi.org/10.1017/S1351324900000139.

Colunga, E. and L. B. Smith. 2005. From the lexicon to expectations about kinds: A role for associative learning. *Psychological Review*, 112(2):347–382. https://doi.org/10.1037/0033-295X.112.2.347.

Diehl Martinez, R., Z. Goriely, H. McGovern, C. Davis, A. Caines, P. Buttery, and L. Beinborn. 2023. CLIMB – Curriculum learning for infant-inspired model building. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*, pages 112–127. https://doi.org/10.18653/v1/2023.conll-babylm.10.

Du Bois, J. W. and R. Englebretson. 2004. Santa Barbara Corpus of Spoken American English. TalkBank. https://doi.org/10.21415/T5VG6X.

Elman, J. L. 1993. Learning and development in neural networks: The importance of starting small. *Cognition*, 48(1):71–99. https://doi.org/10.1016/0010-0277(93)90058-4.

Feinman, R. and B. M. Lake. 2018. Learning inductive biases with simple neural networks. In *Proceedings of the 40th Annual Meeting of the Cognitive Science Society*, pages 1657–1662.

Feng, S. Y., N. D. Goodman, and M. C. Frank. 2024. Is child-directed speech effective training data for language models? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 22055–22071. https://doi.org/10.18653/v1/2024.emnlp-main.1231.

Fenson, L., V. A. Marchman, D. J. Thal, P. S. Dale, J. S. Reznick, and E. Bates. 2007. *MacArthur-Bates Communicative Development Inventories: User's Guide and Technical Manual*, 2nd edition. Paul H. Brookes.

Gao, L., S. Biderman, S. Black, L. Golding, T. Hoppe, C. Foster, J. Phang, H. He, A. Thite, N. Nabeshima, S. Presser, and C. Leahy. 2020. The Pile: An 800GB dataset of diverse text for language modeling. ArXiv:2101.00027.

Geirhos, R., P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel. 2019. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *The Seventh International Conference on Learning Representations*.

Gershkoff-Stowe, L. and L. B. Smith. 2004. Shape and the first hundred nouns. *Child Development*, 75(4):1098–1114. https://doi.org/10.1111/j.1467-8624.2004.00728.x.

Goodman, N. 1955. *Fact, Fiction, and Forecast*. Harvard University Press.

Grattafiori, A., A. Dubey, A. Jauhri, A. Pandey, A. Kadian, et al. 2024. The Llama 3 herd of models. ArXiv:2407.21783.

Haga, A., A. Fukatsu, M. Oba, A. Bisazza, and Y. Oseki. 2024. BabyLM Challenge: Exploring the effect of variation sets on language model training efficiency. In *The 2nd BabyLM Challenge at the 28th Conference on Computational Natural Language Learning*, pages 252–261. https://aclanthology.org/2024.conll-babylm.23/.

Holm, S. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Hu, M. Y., A. Mueller, C. Ross, A. Williams, T. Linzen, C. Zhuang, R. Cotterell, L. Choshen, A. Warstadt, and E. G. Wilcox. 2024. Findings of the Second BabyLM Challenge: Sample-efficient pretraining on developmentally plausible corpora. In *The 2nd BabyLM Challenge at the 28th Conference on Computational Natural Language Learning*, pages 1–21. https://aclanthology.org/2024.conll-babylm.1/.

Huebner, P. A., E. Sulem, C. Fisher, and D. Roth. 2021. BabyBERTa: Learning more grammar with small-scale child-directed language. In *Proceedings of the 25th Conference on Computational Natural Language Learning*, pages 624–646. https://doi.org/10.18653/v1/2021.conll-1.49.

Kallini, J., I. Papadimitriou, R. Futrell, K. Mahowald, and C. Potts. 2024. Mission: Impossible language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14691–14714. https://doi.org/10.18653/v1/2024.acl-long.787.

Kemp, C., A. Perfors, and J. B. Tenenbaum. 2007. Learning overhypotheses with hierarchical Bayesian models. *Developmental Science*, 10(3):307–321. https://doi.org/10.1111/j.1467-7687.2007.00585.x.

Kuhn, R. and R. De Mori. 1990. A cache-based natural language model for speech recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12(6):570–583. https://doi.org/10.1109/34.56193.

Küntay, A. and D. I. Slobin. 1996. Listening to a Turkish mother: Some puzzles for acquisition. In D. I. Slobin, J. Gerhardt, A. Kyratzis, and J. Guo, editors, *Social Interaction, Social Context, and Language: Essays in Honor of Susan Ervin-Tripp*. Lawrence Erlbaum, pages 265–286.

Lakens, D. 2017. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362. https://doi.org/10.1177/1948550617697177.

Landau, B., L. B. Smith, and S. S. Jones. 1988. The importance of shape in early lexical learning. *Cognitive Development*, 3(3):299–321. https://doi.org/10.1016/0885-2014(88)90014-7.

MacWhinney, B. 2000. *The CHILDES Project: Tools for Analyzing Talk*, 3rd edition. Lawrence Erlbaum.

Madsen, R. E., D. Kauchak, and C. Elkan. 2005. Modeling word burstiness using the Dirichlet distribution. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 545–552. https://doi.org/10.1145/1102351.1102420.

McCoy, R. T., R. Frank, and T. Linzen. 2020. Does syntax need to grow on trees? Sources of hierarchical inductive bias in sequence-to-sequence networks. *Transactions of the Association for Computational Linguistics*, 8:125–140. https://doi.org/10.1162/tacl_a_00304.

Misra, K., J. Rayz, and A. Ettinger. 2023. COMPS: Conceptual minimal pair sentences for testing robust property knowledge and its inheritance in pre-trained language models. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2928–2949. https://doi.org/10.18653/v1/2023.eacl-main.213.

Olsson, C., N. Elhage, N. Nanda, N. Joseph, N. DasSarma, T. Henighan, B. Mann, A. Askell, Y. Bai, A. Chen, et al. 2022. In-context learning and induction heads. ArXiv:2209.11895.

Onnis, L., H. R. Waterfall, and S. Edelman. 2008. Learn locally, act globally: Learning language from variation set cues. *Cognition*, 109(3):423–430. https://doi.org/10.1016/j.cognition.2008.10.004.

Perry, L. K., L. K. Samuelson, L. M. Malloy, and R. N. Schiffer. 2010. Learn locally, think globally: Exemplar variability supports higher-order generalization and word learning. *Psychological Science*, 21(12):1894–1902. https://doi.org/10.1177/0956797610389189.

Prystawski, B., M. Y. Li, and N. D. Goodman. 2023. Why think step by step? Reasoning emerges from the locality of experience. In *Advances in Neural Information Processing Systems 36*, pages 70926–70947. https://doi.org/10.52202/075280-3107.

Qwen Team. 2024. Qwen2.5 technical report. ArXiv:2412.15115.

Reddy, G. 2024. The mechanistic basis of data dependence and abrupt learning in an in-context classification task. In *The Twelfth International Conference on Learning Representations*.

Ritter, S., D. G. T. Barrett, A. Santoro, and M. M. Botvinick. 2017. Cognitive psychology for deep neural networks: A shape bias case study. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of Proceedings of Machine Learning Research, pages 2940–2949.

Rohde, D. L. T. and D. C. Plaut. 1999. Language acquisition in the absence of explicit negative evidence: How important is starting small? *Cognition*, 72(1):67–109. https://doi.org/10.1016/S0010-0277(99)00031-1.

van Schijndel, M. and T. Linzen. 2018. A neural model of adaptation in reading. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 4704–4710. https://doi.org/10.18653/v1/D18-1499.

Schoenegger, L., L. Thoma, T. Blevins, and B. Roth. 2025. Influence-driven curriculum learning for pre-training on limited data. In *Proceedings of the First BabyLM Workshop*, pages 356–379. https://doi.org/10.18653/v1/2025.babylm-main.26.

Schwab, J. F. and C. Lew-Williams. 2016. Repetition across successive sentences facilitates young children's word learning. *Developmental Psychology*, 52(6):879–886. https://doi.org/10.1037/dev0000125.

Shi, W., S. Min, M. Lomeli, C. Zhou, M. Li, G. Szilvasy, R. James, X. V. Lin, N. A. Smith, L. Zettlemoyer, S. Yih, and M. Lewis. 2024. In-context pretraining: Language modeling beyond document boundaries. In *The Twelfth International Conference on Learning Representations*.

Sinclair, A., J. Jumelet, W. Zuidema, and R. Fernández. 2022. Structural persistence in language models: Priming as a window into abstract language representations. *Transactions of the Association for Computational Linguistics*, 10:1031–1050. https://doi.org/10.1162/tacl_a_00504.

Singh, A. K., S. C. Y. Chan, T. Moskovitz, E. Grant, A. M. Saxe, and F. Hill. 2023. The transient nature of emergent in-context learning in transformers. In *Advances in Neural Information Processing Systems 36*, pages 27801–27819.

Slone, L. K., D. H. Abney, L. B. Smith, and C. Yu. 2023. The temporal structure of parent talk to toddlers about objects. *Cognition*, 230:105266. https://doi.org/10.1016/j.cognition.2022.105266.

Smith, L. B., S. S. Jones, B. Landau, L. Gershkoff-Stowe, and L. Samuelson. 2002. Object name learning provides on-the-job training for attention. *Psychological Science*, 13(1):13–19. https://doi.org/10.1111/1467-9280.00403.

Thrush, T., E. Wilcox, and R. Levy. 2020. Investigating novel verb learning in BERT: Selectional preference classes and alternation-based syntactic generalization. In *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pages 265–275. https://doi.org/10.18653/v1/2020.blackboxnlp-1.25.

Vlach, H. A., C. M. Sandhofer, and N. Kornell. 2008. The spacing effect in children's memory and category induction. *Cognition*, 109:163–167. https://doi.org/10.1016/j.cognition.2008.07.013.

Vong, W. K., W. Wang, A. E. Orhan, and B. M. Lake. 2024. Grounded language acquisition through the eyes and ears of a single child. *Science*, 383(6682):504–511. https://doi.org/10.1126/science.adi1374.

van der Wal, O., P. Lesci, M. Müller-Eberstein, N. Saphra, H. Schoelkopf, W. Zuidema, and S. Biderman. 2025. PolyPythias: Stability and outliers across fifty language model pre-training runs. In *The Thirteenth International Conference on Learning Representations*.

Wang, W., G. Jiang, T. Linzen, and B. M. Lake. 2025. Rapid word learning through meta in-context learning. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 32038–32073.

https://doi.org/10.18653/v1/2025.emnlp-main.1631.
Wang, W., W. K. Vong, N. Kim, and B. M. Lake. 2023. Finding structure in one child's linguistic experience. *Cognitive Science*, 47(6):e13305. https://doi.org/10.1111/cogs.13305.
Warstadt, A., A. Mueller, L. Choshen, E. Wilcox, C. Zhuang, J. Ciro, R. Mosquera, B. Paranjabe, A. Williams, T. Linzen, and R. Cotterell. 2023. Findings of the BabyLM Challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*, pages 1–34. https://doi.org/10.18653/v1/2023.conll-babylm.1.
White, J. C. and R. Cotterell. 2021. Examining the inductive bias of neural language models with artificial languages. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 454–463. https://doi.org/10.18653/v1/2021.acl-long.38.
Yedetore, A., T. Linzen, R. Frank, and R. T. McCoy. 2023. How poor is the stimulus? Evaluating hierarchical generalization in neural networks trained on child-directed speech. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9370–9393. https://doi.org/10.18653/v1/2023.acl-long.521.
Zhao, Y., Y. Qu, K. Staniszewski, S. Tworkowski, W. Liu, P. Miłoś, Y. Wu, and P. Minervini. 2024. Analysing the impact of sequence composition on language model pre-training. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7897–7912. https://doi.org/10.18653/v1/2024.acl-long.427.

Appendix A: Registration record

All registrations are in one OSF project, osf.io/qj9hb, and each states what was known when it was written. Table A1 lists them, Table A2 gives the label each registration uses for each row of Table A4, and Table A4 gives every registered hypothesis and its result.

The loading correction. The model code ties the input embedding to the output layer. On the graphics card used for training (through DirectML) the tie was lost, so every Study 1 model was trained and saved with two separate matrices. The first version of the scoring code rebuilt each model with the tie in place before loading its weights, so the output matrix overwrote the input embedding. We found this on 12 October 2026, fixed the loader, and rescored all 315 Study 1 models then trained, 461 intermediate checkpoints of the random and grouped Regular models and the trained-noun items, with the same items. A check on eight randomly chosen models confirmed the fix: mean loss on each model's own training corpus was 0.07 to 0.16 nats per token with the corrected loader, close to the last logged training loss (0.08 to 0.23; dropout is off at test), and 0.58 to 1.97 with the old one. Studies 2 and 3 were not affected. The old scores are kept in the repository.

Table A1
Registrations. Each was written before the models it concerns were trained or scored, or before the transcripts were read.

Registration	Written	Rows of Table A4	Known when it was written
Original	2 Mar 2026	Corpora, models and controls (Appendix D)	Nothing trained
Amendment 1	7 Oct 2026	1, 2, 3, 6, 16	Random-order models trained and scored on an earlier test, with the old loader; no grouped model trained
Amendment 2	7 Oct 2026	7 (texture added as a co-primary test)	No grouped model trained
Amendment 3	8 Oct 2026	23, 24, 25	Study 1 results for rows 1 to 3, 6, 7 and 16 (old loader); no pretrained model scored
Amendment 4	8 Oct 2026	20, 21	Studies 1 and 3 results; no transcript read
Amendment 5	9 Oct 2026	4, 5, 8, 9, 17, 18, 19, 22	All earlier results (Study 1 with the old loader); four reviews of a draft; five exploratory checks, including the confound of the noun with another
Amendment 6	11 Oct 2026	11, 12, 13	Amendment 5 results (old loader); a prediction for the new models fitted to them
Amendment 7	12 Oct 2026	10, 14, 15	All results, with corrected scores
Amendment 8	15 Oct 2026	26 to 32	All results, with corrected scores; three rounds of reviews, which noted that fixed-file models memorised their files and that the novel-noun test was uninformative; no fresh-text model trained

Table A2
Rows of Table A4 and the labels used in the registrations.

Row	Label	Row	Label	Row	Label	Row	Label
1	H4b	9	H13b	17	H14a	25	H8
2	H1b	10	H22	18	H14b	26	H23
3	H2b	11	H17	19	H15	27	H24
4	H11	12	H18	20	H9	28	H25
5	H12	13	H19	21	H10	29	H26
6	H5b colour	14	H20	22	H16	30	H27
7	H5b texture	15	H21	23	H6	31	H28
8	H13a	16	H3b	24	H7	32	H29

Table A3
Study 1 registered tests before and after the correction (shape-over-colour index, nats). Rows 10, 14 and 15 were registered and scored after the correction.

Row	Old loader	Corrected	Verdict before → after
1	1.36 vs 0.18	6.49 vs 0.47	Supported → supported
2	0.18	0.47	Not supported → not supported
3	0.18 vs 0.09	0.47 vs −0.25	Not supported → supported
4	difference 1.45	difference 6.93	Supported → supported
5	difference 1.17	difference 2.76	Supported → supported
6	1.36 vs 0.80	6.49 vs 2.12	Supported → supported
7	1.10 vs −0.04	5.36 vs 0.46	Supported → supported
8	slope 0.29	slope 1.63	Supported → supported
9	0.24 vs 0.18	1.43 vs 0.47	Not supported → supported
11	0.21 vs 0.18	0.50 vs 0.47	Not supported → not supported
12	0.29 vs 0.24	0.80 vs 1.43	Not supported → not supported (reversed)
13	0.21 vs −0.08	0.50 vs 0.10	Supported → supported
16	difference 0.0007	difference 0.005	Equivalent to zero → equivalent to zero

Table A4
All registered hypotheses and results; registered secondary analyses are reported in the text. Rows 1 to 16 concern fixed-file models and rows 26 to 32 fresh-text models. Seed-level tests average the three sizes within each of five seeds (t-based 95% intervals, two-sided; tests one-sided unless marked, 4 degrees of freedom), so a one-sided test can be below .05 while the two-sided interval includes zero (row 13). The registered p is the test named in the registration, one-sided unless marked, Holm-corrected within its registration; for rows 26 to 32 it is the seed-level test, corrected within rows 26 to 29 and within rows 30 to 32. Corpus intervals resample sources and then documents; Pythia and BabyLM intervals resample nouns.

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
	1. Does grouping teach the fixed feature?					
1	Grouped order gives a higher index than random order	6.49 vs 0.47	6.02 [5.24, 6.79]	$< .001$	$< .001$	Supported
2	Random order gives an index above zero	0.47	0.47 [−0.61, 1.56]	.15	.061	Not supported
3	Random order: Regular above Mixed-slot	0.47 vs −0.25	0.73 [−0.69, 2.15]	.11	.032	Supported by the registered test, not at the seed level
4	Grouped: Regular above No stable feature	6.49 vs −0.44	6.93 [5.47, 8.40]	$< .001$	$< .001$	Supported
5	Grouped: Colour stable below No stable feature	−3.20 vs −0.44	2.76 [1.99, 3.52]	$< .001$	$< .001$	Supported
6	Grouped: Regular above Mixed-slot, shape over colour	6.49 vs 2.12	4.37 [1.98, 6.76]	.004	$< .001$	Supported
7	Grouped: Regular above Mixed-slot, shape over texture	5.36 vs 0.46	4.89 [2.69, 7.10]	.002	$< .001$	Supported
8	The index rises with log B across six orders	Slope 1.63 (SE 0.10)	1.63 [1.46, 1.80]	$< .001$	$< .001$	Supported
9	B = 3.6 above random order	1.43 vs 0.47	0.96 [0.63, 1.28]	$< .001$	$< .001$	Supported
10	The same, Objects corpora, different-object runs	1.18 vs 0.22	0.96 [0.37, 1.54]	.005	$< .001$	Supported
26	Fresh text: grouped order gives a higher index than random order	9.52 vs −1.56	11.08 [9.57, 12.59]	$< .001$	$< .001$	Supported
27	Fresh text: B = 3.5 above random order	1.28 vs −1.56	2.84 [1.41, 4.27]	.003	.005	Supported
28	Fresh text, grouped: Regular above No stable feature	9.52 vs −3.98	13.50 [11.48, 15.52]	$< .001$	$< .001$	Supported
	2. What kind of grouping helps?					
11	512-token window above 128, random order	0.50 vs 0.47	0.03 [−0.44, 0.50]	.44	.87	Not supported
12	512-token window above 128, B = 3.6	0.80 vs 1.43	−0.63 [−0.90, −0.37]	.003 (two-sided)	1.00	Not supported; reliably reversed
13	512-token window: Regular above No stable feature	0.50 vs 0.10	0.40 [−0.04, 0.85]	.032	.045	Supported; weak
14	Grouped: different-object above same-object runs	5.65 vs 2.99	2.66 [1.57, 3.75]	.001	$< .001$	Supported

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
15	B = 3.6: different-object above same-object runs	1.18 vs 0.45	0.73 [0.15, 1.32]	.013	$< .001$	Supported
29	Fresh text, grouped: different-object above same-object runs 3. Is the expectation tied to the noun?	7.60 vs 5.87	1.72 [0.54, 2.90]	.008	.008	Supported
16	Fixed files, random order: index higher after the same noun (novel nouns)	Difference 0.005; equivalent to zero (p = .001), but the novel nouns were untrained (Section 3.5)			.46	Not supported
30	Shape reassigned, grouped: noun-specific index above zero, novel nouns	0.37	0.37 [0.24, 0.49]	$< .001$	.001	Supported
31	The same, above grouped Regular (fresh text)	0.37 vs 0.12	0.25 [0.10, 0.39]	.005	.005	Supported
32	Shape reassigned, grouped: noun-specific index above zero, nouns from training	11.24	11.24 [10.44, 12.03]	$< .001$	$< .001$	Supported
17	Pythia: noun-specific index rises with size (wording matched)	$\rho = -.42$			1.00	Not supported
18	Pythia 410M to 12B: noun-specific index above zero (wording matched)	−0.15 (SE 0.12)			1.00	Not supported
19	Pythia: 410M above 160M across ten training seeds 4. What do natural input and pretrained models show?	0.15 vs −0.15			.012	Supported
20	Speech to children: B above 1	3.57 [3.13, 4.30]			$< .001$	Supported
21	Speech to children above written text	3.57 vs 1.43; difference 2.14 [1.55, 2.91]			$< .001$	Supported
22	Speech to children differs from adult conversation (two-sided)	3.57 vs 2.33; difference 1.24 [0.51, 2.03]			$< .001$	Supported; higher
23	Pythia: index rises with size	$\rho = .92$			.003	Supported
24	Pythia: index rises over training	Mean within-size $\rho = -.34$			1.00	Not supported
25	BabyLM baselines (10M, 100M words): index above zero	0.47 [−1.05, 1.93]; 0.37 [−0.04, 0.73]			.52; .12	Not supported

Fresh text and shape reassignment. Rows 26 to 32 were registered on 15 October 2026, after reviews of the manuscript had noted that fixed-file models memorised their files and that the test with novel nouns was uninformative, and before any fresh-text model was trained. The corpora were built, and their checksums registered, before training. Amendment 8 names the conditions by their folder names (for example reassigned_grouped).

Appendix B: Deviations from the registrations

1. Two registered analyses are not reported here: the Bayesian reference model of Amendment 1, which is in the repository, and the prediction of Amendment 6 for the 512-token models, which had been fitted to scores later corrected (Appendix A).
2. Added after registration: the seed-level tests for rows 1 to 3 and 6 to 8, the seed-level intervals, and intervals for the differences between natural corpora (from the registered two-stage bootstrap). Effective clustering was computed before it was registered as the scale for Figure 2 and is reported as exploratory (Table 3).
3. The two models of about 70 billion parameters were scored with Apple's MLX library, because PyTorch crashed the machine; on smaller models the two differed by 0.003 to 0.010 nats on average.
4. In the hand coding of Study 2, a rule for labels not used for a physical object was added during coding and applied by both coders, and one pair of the reliability subset was discussed with an AI assistant and left out of the agreement check.

Smaller technical changes (software versions, download addresses, a stale file in one public model repository) are listed in the repository.

Appendix C: Model and training settings

Table C1
AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up then cosine decay, dropout 0.1, untied input and output embeddings. Fixed-file models read windows starting at random positions in the corpus; fresh-text models read non-overlapping windows, each at most once, in random order (the 3.4M and 25.6M models used about 60% of the windows available, the 11.0M models 94 to 96%). Byte-pair tokeniser with 351 entries for the Regular corpora and 314 to 339 for the other corpus families; every artificial word is one token. Seeds 42, 123, 456, 789 and 1001 each set the corpus, the initial weights and the batch order. Checkpoints were saved every 500 steps. Training ran on one Windows PC through DirectML.

Setting	3.4M	11.0M	25.6M
Layers, heads, width	4, 4, 256	6, 8, 384	8, 8, 512
Feed-forward width	1,024	1,536	2,048
Steps	5,000	8,000	10,000
Batch (128-token windows)	32	32	16
Batch (512-token windows)	8	8	4
Peak learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}
Warm-up steps	200	400	500

Table C2
Memorisation. Mean next-token loss (nats per token) on 300 windows from each model's own training text and from a new arrangement of the same sentences (100 further passes, 27 for the Objects corpora, never used in training); 15 models per cell. At the seed level, the gap between the two losses was larger for fixed-file than for fresh-text models by 0.57 to 1.16 nats, with every interval excluding zero.

Corpus	Fixed file: own file	Fixed file: new arrangement	Fresh text: own text	Fresh text: new arrangement
Regular, random	0.11	1.25	0.55	0.55
Regular, B 3.6 (fresh: 3.5)	0.10	1.26	0.54	0.54
Regular, grouped	0.09	0.85	0.35	0.35
No stable feature, grouped	0.09	1.00	0.39	0.39
Objects, same-object runs, grouped	0.08	0.66	0.28	0.28
Objects, different-object runs, grouped	0.09	0.76	0.32	0.32

Appendix D: Further content controls

Six further corpora, each trained in random order only, change one thing about Regular. **No kind label**: the nouns are removed, with no other marker of the kind. **Paraphrased**: the nouns are removed and replaced by randomly chosen category markers. **Weak label**: nouns appear in 25% of sentences and category markers in the rest. **Frequency matched**: the same feature words as often as in Regular, with the fixed feature of each kind drawn at random from shape, texture or none. **Added noise**: Regular with 7% of exemplars given a wrong fixed feature. **Feature swap**: 16 kinds keep their shape and 16 keep their texture.

Table D1
Further controls, random order, 15 models each. In random order every index is small, so these corpora say little about what grouping would do with them.

Corpus	Shape over colour, mean (SD)	Shape over texture	Noun-specific index
Regular (for comparison)	0.47 (0.90)	0.51	0.005
No kind label	0.06 (0.49)	0.44	−0.004
Paraphrased	0.41 (0.44)	0.60	−0.025
Weak label	0.09 (0.50)	−0.52	0.008
Frequency matched	−0.22 (0.63)	−0.40	−0.033
Added noise	−0.43 (0.67)	0.44	−0.038
Feature swap	0.69 (0.77)	0.60	−0.017

Appendix E: Robustness

Table E1
p values for the Study 1 rows of Table A4 under paired tests at the seed level and over the 15 matched models (one-sided unless marked).

Row	Seed level (5 seeds)	Matched models (15)
1	$< .001$	$< .001$
2	.15	.031
3	.11	.014
4	$< .001$	$< .001$
5	$< .001$	$< .001$
6	.004	$< .001$
7	.002	$< .001$
8	$< .001$	
9	$< .001$	$< .001$
10	.005	$< .001$
11	.44	.44
12	.003 (two-sided)	.003 (two-sided)
13	.032	.015
14	.001	$< .001$
15	.013	$< .001$
26	$< .001$	$< .001$
27	.003	$< .001$
28	$< .001$	$< .001$
29	.008	$< .001$
30	$< .001$	$< .001$
31	.005	.010
32	$< .001$	$< .001$

Table E2
Shape-over-colour index by model size (one exemplar), and with two or four exemplars in the context that share their shape (all sizes). Rows marked fresh are fresh-text models; the rest are fixed-file models. In grouped Regular models, two or four exemplars lowered the index: a run of a kind has four sentences, so a fifth sentence about the same kind is rare (Section 3.5). In grouped No stable feature and Colour stable models, where shape varies within a kind, more exemplars sharing a shape lowered the index (Δ shape −0.64, −1.93 and −3.42 with one, two and four exemplars in grouped No stable feature models), for which we have no tested account; in random-order Regular models, too, the index was lower with two or four exemplars than with one.

Condition	3.4M	11.0M	25.6M	Two exemplars	Four exemplars
Regular, random, fresh	−1.00	−0.87	−2.81	−1.83	−2.19
Regular, B 3.5, fresh	1.37	2.27	0.21	2.05	−0.53
Regular, grouped, fresh	7.43	12.57	8.56	7.88	4.24
No stable feature, grouped, fresh	−1.21	−4.80	−5.92	−5.41	−7.24
Objects, same-object, grouped, fresh	8.21	4.60	4.81	3.86	0.62
Objects, different-object, grouped, fresh	9.96	6.29	6.54	5.78	2.13
Reassigned, random, fresh	−0.06	0.33	0.35	0.59	0.23
Reassigned, grouped, fresh	6.52	14.05	8.85	9.62	8.26
Regular, random	0.33	0.71	0.38	−0.64	−0.10
Regular, B 1.4	0.73	1.05	0.70	−0.01	0.32
Regular, B 3.6	0.97	2.05	1.27	0.60	1.06
Regular, B 7	1.85	2.89	1.45	1.23	1.15
Regular, B 12	3.18	4.53	2.88	2.85	2.43
Regular, grouped	6.41	6.83	6.23	5.24	4.30
Mixed-slot, random	−0.11	−0.45	−0.21	0.29	0.29
Mixed-slot, grouped	2.24	2.35	1.76	2.60	2.04
No stable feature, random	−0.10	−0.33	−0.12	−0.17	−0.44
No stable feature, grouped	−0.41	−0.28	−0.63	−1.77	−3.26
Colour stable, random	−0.04	−0.70	−0.51	−0.76	−0.76
Colour stable, grouped	−2.11	−3.72	−3.77	−4.50	−5.34
Regular, random, 512	0.30	0.51	0.70	0.51	0.32
Regular, B 3.6, 512	0.31	1.02	1.06	0.93	0.82
No stable feature, random, 512	0.21	0.04	0.05	−0.10	−0.23
Objects, random	0.36	0.44	−0.13	0.38	1.02
Objects, same-object, B 3.6	0.22	0.98	0.14	0.32	1.22
Objects, different-object, B 3.6	0.83	1.75	0.96	1.49	2.18
Objects, same-object, grouped	3.00	3.33	2.62	1.30	1.34
Objects, different-object, grouped	6.18	5.43	5.34	4.54	4.14

Noun-specific index. At the seed level, the noun-specific index was equivalent to zero (two one-sided tests, ± 0.2 nats) in 25 of the 26 fixed-file conditions; the exception was same-object runs at B = 3.6 (−0.11, p = .058). No condition exceeded 0.15 nats in absolute value. These nulls are uninformative, because the models did not track novel nouns (Section 3.5). As a further check, for the 60 models in the random and grouped Regular and grouped Objects conditions we compared the log probability of the same novel noun after the context and the word A with the mean over the other seven novel nouns. The difference was no more than 0.032 nats in absolute value in every model, and positive on about half the items. Because the novel nouns were never training targets, this check also depends on output weights that training only pushed down, so it is weaker evidence than the trained-noun contrast in Section 3.5.

Correction across all tests. Holm correction was applied within each registered family of tests, as registered. Every row of Table A4 with a Holm-adjusted p below .001

has an unadjusted p below .001, and so stays below .05 under Holm correction across all 32 tests.

What the corpora reward. For every pair of consecutive sentences in each training corpus (five seeds pooled), Table E3 gives the share about the same kind and how often the second sentence repeats the first one's shape or colour. The noun-blind reference index is D(shape) − D(colour), where D = ln p − ln((1 − p)/9) and p is the share of pairs that repeat that feature, with ten values per dimension; it is what a learner that predicts each feature from the previous sentence alone would show on a one-exemplar item. The noun-aware reference computes the same index separately for pairs about the same kind and pairs about different kinds (add-0.5 smoothing).

Table E3
What each training corpus says about the next sentence, and the models' shape-over-colour index for comparison. Rows marked fresh describe fresh text, counted on 100 held-out passes arranged by the same rule (27 for the Objects corpora); the rest describe the fixed files. All values in nats except the shares. Code is in the repository.

Corpus	Next sentence same kind	Shape repeats, same kind	Shape repeats, other kind	Colour repeats, same kind	Noun-blind reference	Noun-aware, same kind	Noun-aware, other kind	Model index
Regular, random, fresh	0.03	1.00	0.07	0.10	−0.01	11.7	−0.35	−1.56
Regular, B 3.5, fresh	0.11	1.00	0.07	0.09	0.65	13.1	−0.37	1.28
Regular, grouped, fresh	0.72	1.00	0.07	0.10	3.25	14.9	−0.38	9.52
No stable feature, grouped, fresh	0.72	0.10	0.10	0.10	0.02	0.0	−0.01	−3.98
Objects, same-object, grouped, fresh	0.76	1.00	0.07	0.99	0.39	8.4	−0.35	5.87
Objects, different-object, grouped, fresh	0.76	1.00	0.07	0.10	3.43	14.9	−0.29	7.60
Reassigned, random, fresh	0.03	1.00	0.10	0.10	0.27	8.2	0.01	0.21
Reassigned, grouped, fresh	0.72	1.00	0.10	0.10	3.30	11.4	−0.02	9.80
Regular, random	0.03	1.00	0.07	0.10	−0.01	10.3	−0.35	0.47
Regular, B 1.4	0.04	1.00	0.07	0.10	0.13	10.8	−0.35	0.83
Regular, B 3.6	0.11	1.00	0.07	0.10	0.70	11.7	−0.31	1.43
Regular, B 7	0.22	1.00	0.07	0.10	1.23	12.4	−0.37	2.06
Regular, B 12	0.38	1.00	0.07	0.09	1.91	13.0	−0.40	3.53
Regular, grouped	0.72	1.00	0.07	0.10	3.25	13.5	−0.33	6.49
No stable feature, grouped	0.72	0.10	0.10	0.10	0.03	0.0	0.06	−0.44
Colour stable, grouped	0.72	0.10	0.10	1.00	−3.26	−12.5	0.38	−3.20
Objects, same-object, B 3.6	0.11	1.00	0.07	0.70	0.19	8.6	−0.35	0.45
Objects, different-object, B 3.6	0.11	1.00	0.07	0.11	0.74	11.6	−0.25	1.18
Objects, same-object, grouped	0.76	1.00	0.07	0.99	0.37	7.0	−0.39	2.99
Objects, different-object, grouped	0.76	1.00	0.07	0.10	3.39	13.5	−0.41	5.65

Table E4
Study 1 results for models trained on fixed files, in nats after one exemplar with the same noun; 15 models per row, 128-token windows unless marked.

Corpus	Order (B)	Δ shape	Δ colour	Shape over colour, mean (SD)	Models above zero	Shape over texture
Regular	Random (0.9)	0.50	0.03	0.47 (0.90)	10 of 15	0.51
Regular	Graded (1.4)	0.77	−0.05	0.83 (0.72)	12	0.81
Regular	Graded (3.6)	1.33	−0.10	1.43 (0.93)	15	1.28
Regular	Graded (7)	1.71	−0.36	2.06 (0.95)	15	1.95
Regular	Graded (12)	2.94	−0.59	3.53 (1.12)	15	3.09
Regular	Grouped (23)	5.36	−1.13	6.49 (1.49)	15	5.36
Mixed-slot	Random	−0.37	−0.12	−0.25 (0.72)	4	−0.54
Mixed-slot	Grouped	0.76	−1.36	2.12 (1.07)	14	0.46
No stable feature	Random	−0.01	0.17	−0.18 (0.39)	5	−0.09
No stable feature	Grouped	−0.64	−0.20	−0.44 (0.54)	4	−1.02
Colour stable	Random	−0.06	0.36	−0.42 (0.70)	4	−0.06
Colour stable	Grouped	−0.81	2.39	−3.20 (1.18)	0	−0.22
Regular	Random (0.9), 512-token window	0.37	−0.13	0.50 (0.62)	12	0.41
Regular	Graded (3.6), 512-token window	0.67	−0.13	0.80 (0.79)	11	0.67
No stable feature	Random, 512-token window	0.07	−0.03	0.10 (0.60)	9	−0.10
Objects	Random (1.0)	0.58	0.36	0.22 (0.43)	11	0.40
Objects	Same-object runs (3.6)	1.34	0.90	0.45 (0.62)	12	1.40
Objects	Different-object runs (3.6)	1.25	0.07	1.18 (0.67)	14	1.13
Objects	Same-object runs (24)	4.63	1.65	2.99 (1.05)	15	4.05
Objects	Different-object runs (24)	5.37	−0.28	5.65 (1.18)	15	5.07

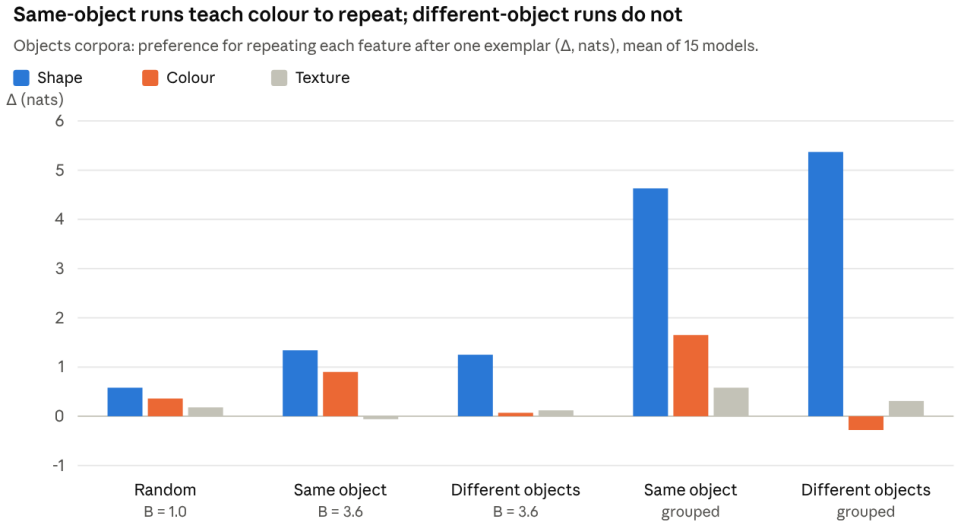


Figure E1
Preference for repeating each feature after one exemplar (Δ, nats; mean of 15 models per condition), in fixed-file models trained on the Objects corpora. Same-object runs raise the preference for repeating colour along with shape; different-object runs raise only shape.

Appendix F: Trained nouns

All Study 1 models except those trained on the further content controls (Appendix D), 420 in all, were also scored on 960 items whose nouns occurred in training, in the format of Section 3.3: 16 context nouns, 10 draws of features, three probed dimensions and two continuation nouns (the same noun or another trained noun). No shape that is the fixed feature of either noun appears in an item.

Table F1
Trained nouns, selected conditions (all conditions in the repository). Rows marked fresh are fresh-text models; the rest are fixed-file models. Except after shape reassignment, the index on trained nouns is closer to zero than on novel nouns. Indices in nats; "models above zero" and p (two-sided one-sample t over 15 models) refer to the noun-specific index. For grouped Regular fixed-file models the registered one-sided test gives p = .008.

Condition	Index, same noun	Noun-specific index	Models above zero	p (two-sided)
Regular, random, fresh	−0.02	−0.09	7 of 15	.59
Regular, B 3.5, fresh	−0.13	−0.15	7	.19
Regular, grouped, fresh	2.00	1.04	10	.009
No stable feature, grouped, fresh	−3.75	−2.60	0	$< .001$
Objects, same-object, grouped, fresh	−0.55	0.05	8	.74
Objects, different-object, grouped, fresh	0.03	0.44	13	$< .001$
Reassigned, random, fresh	0.14	−0.07	5	.51
Reassigned, grouped, fresh	11.65	11.24	15	$< .001$
Regular, random	0.07	0.01	9	.97
Regular, B 3.6	0.36	0.12	10	.44
Regular, B 7	0.49	0.18	8	.29
Regular, B 12	1.07	0.06	8	.64
Regular, grouped	1.03	0.30	10	.015
Objects, same-object, grouped	−1.53	−1.06	0	$< .001$
Objects, different-object, grouped	0.33	0.14	10	.26

Appendix G: Materials for Studies 2 and 3

Object nouns (Study 2). ball, balloon, block, book, crayon, doll, puzzle, toy; airplane, bicycle, boat, bus, car, motorcycle, stroller, tractor, train, truck; apple, banana, carrot, cookie, cracker, egg, grape, raisin, sandwich; boot, button, coat, diaper, hat, jacket, mitten, necklace, pajamas, pants, shirt, shoe, sock, sweater, zipper; basket, blanket, bottle, bowl, box, broom, clock, cup, dish, fork, hammer, key, lamp, pencil, pillow, plate, purse, scissors, spoon, telephone, toothbrush, towel; bed, chair, couch, crib, table, television.

Coding rules (Study 2). Each pair was shown with the utterance before it, the stretch between the two mentions and the utterance after, without its source. *Same*: both mentions refer to the same physical object. *Different*: two or more distinct objects of the kind (another, other, numbers, plurals that count objects, descriptions that cannot fit one object). *Generic*: at least one mention refers to the kind in general. *Unclear*: the words do not settle it, including labels not used for a physical object. The second coder coded a random 25% of pairs (seed 2031). The full rubric and the second coder's prompt are in the repository.

Feature words (Study 3). Word list 1: shape *pointed, oval, circular, curved, rectangular, triangular, spherical, cylindrical*; colour *grey, purple, amber, violet, scarlet, crimson, maroon, turquoise*; texture *smooth, shiny, sticky, fuzzy, fluffy, furry, glossy, silky*. Word list 2: shape *oblong, hexagonal, conical, tapered, elliptical, tubular, cubic, rounded*; colour *magenta, lilac, teal, beige, indigo, lavender, ivory, coral*; texture *velvety, waxy, prickly, bumpy, gritty, coarse, slippery, rough*. Mean Zipf frequencies for list 1 are 3.75 (shape), 3.82 (colour) and 3.70 (texture).

Nouns (Study 3). Original items: snerg, twick, vulf, wazz, glorp, yeff, zull, kleb. Added sets: these and brell, sprock, fendle, plim, skeb, tulver, grisp, vosh.

Item counts (Study 3). Original items: 576 one-exemplar items (8 nouns, 12 draws, 3 dimensions, 2 continuation nouns) and 2,304 with two or four exemplars sharing either shape or colour, 2,880 in all. Each added set has 1,152 one-exemplar items.

Checkpoints (Study 3). Pythia 14M, 70M, 160M, 410M, 1B, 1.4B and 2.8B at steps 0, 4, 16, 64, 256, 1,000, 4,000, 16,000, 64,000 and 143,000 (2.8B without step 64,000); Pythia 6.9B and 12B at steps 64, 1,000, 4,000, 8,000, 12,000, 16,000, 32,000 and 143,000; Pythia 410M to 2.8B also at every 1,000 steps from 2,000 to 15,000 and at 18,000, 20,000, 24,000 and 32,000; nine further training seeds of Pythia 14M, 31M, 70M, 160M and 410M (and the original 31M model) at steps 4,000, 16,000, 64,000 and 143,000; the BabyLM GPT-2 baselines and Qwen2.5 72B and Llama 3.1 70B at their released weights. Exact revisions are listed in the repository.

Appendix H: Data and code

All code, corpora, test items, model scores, transcript-level counts, hand codes (without transcript excerpts) and analysis outputs are at `github.com/synthiumjp/overhypothesis`. REPRODUCE.md gives the command that regenerates each table and figure, with MD5 checksums for the fixed inputs. The final weights and training logs of the 510 trained Study 1 models are at `huggingface.co/synthiumjp/overhypothesis-study1-models`. The pretrained models are public at the revisions listed. CHILDES and CABank transcripts are available from TalkBank under its terms and are not redistributed. Study 2 is a secondary analysis of these archived, de-identified transcripts and of licensed telephone corpora, used under their terms; no data were collected from people for this paper, and no transcript excerpts are reproduced.